

STAT8561 - Linear Statistical Analysis I

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Preface

The topic of this course includes statistical inference, Multivariate normal distribution, distribution of quadratic forms, linear models, regression models and experimental design models.

Prerequisites

Math 4751/6751 Mathematical Statistics I and Math 4752/6752 Mathematics and Statistics II.

Instructor

Chi-Kuang Yeh, Assistant Professor in the [Department of Mathematics and Statistics, Georgia State University](#).

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Office Hour

9:00 - 12:00 on Wednesday

Grade Distribution

- Homework – 40%
- Exam 1 – 20%
- Exam 2 – 20%
- Final – 20%
- Attendance – *7%

Assignment

- A1, TBA

Exam

- TBA

Topics and Corresponding Lectures

Those chapters are based on the lecture notes. This part will be updated frequently.

Status	Chapter	Topic	Lecture
		Welcome and Overview	1
	Ch. 1	Math Notation	2–

Recommended Textbooks

- Ding, Peng. *Linear Model and Extensions*. CRC Press, 2027. [arXiv link](#)
- Montgomery, Douglas C., Elizabeth A. Peck, and G. Geoffrey Vining. *Introduction to Linear Regression Analysis*. John Wiley & Sons, 2021. (Montgomery et al. 2021)
- Seber, George AF, and Alan J. Lee. *Linear Regression Analysis*. John Wiley & Sons, 2003. (Seber and Lee 2003)

Side Readings

- Montgomery, Douglas C. (2017). *Design and Analysis of Experiments*. John Wiley & Sons. (Montgomery 2017)

Introduction

This is an introduction to the course for linear statistical analysis. It will give you an overview of what we will be covering in the course and how to get the most out of it. We will be covering a wide range of topics in linear statistical analysis,

- simple linear regression
- multiple linear regression
- generalized linear models
- mixed effects models
- quantile regression.

We will also be discussing the assumptions underlying these models and how to check them.

The course will be structured around lectures, homework assignments, a midterm and a final.

To get the most out of this course, it is important to attend all lectures and complete all homework assignments. It is also important to ask questions and participate in class discussions. The more you engage with the material, the more you will learn.

I am looking forward to a great semester and I hope you are too!

To illustrate the concepts we will be covering in this course, let's consider a simple example. Suppose we have a data set of heights and weights of individuals. We want to understand the relationship between height and weight, and we can use linear regression to model this relationship.

```
# Load necessary libraries
library(ggplot2)
library(dplyr)
# Create a sample data set

set.seed(8561)

theme_set(theme_minimal())

n <- 100
heights <- rnorm(n, mean = 170, sd = 10)
weights <- 0.5 * heights + rnorm(n, mean = 0, sd
```

```
= 5)
df <- data.frame(heights, weights)
head(df, 5)
```

```
      heights  weights
1 179.2812 88.01599
2 168.0474 76.10017
3 178.3029 83.46016
4 178.7006 82.01763
5 172.6678 92.84592
```

```
# Fit a linear regression model
model <- lm(weights ~ heights, data = df)
# Summarize the model
summary(model)
```

Call:

```
lm(formula = weights ~ heights, data = df)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-12.3303  -3.8998  -0.2191   3.2380  12.3478
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.92852     8.22226  -0.235    0.815
heights      0.50907     0.04857  10.481 <2e-16 ***
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 5.251 on 98 degrees of freedom

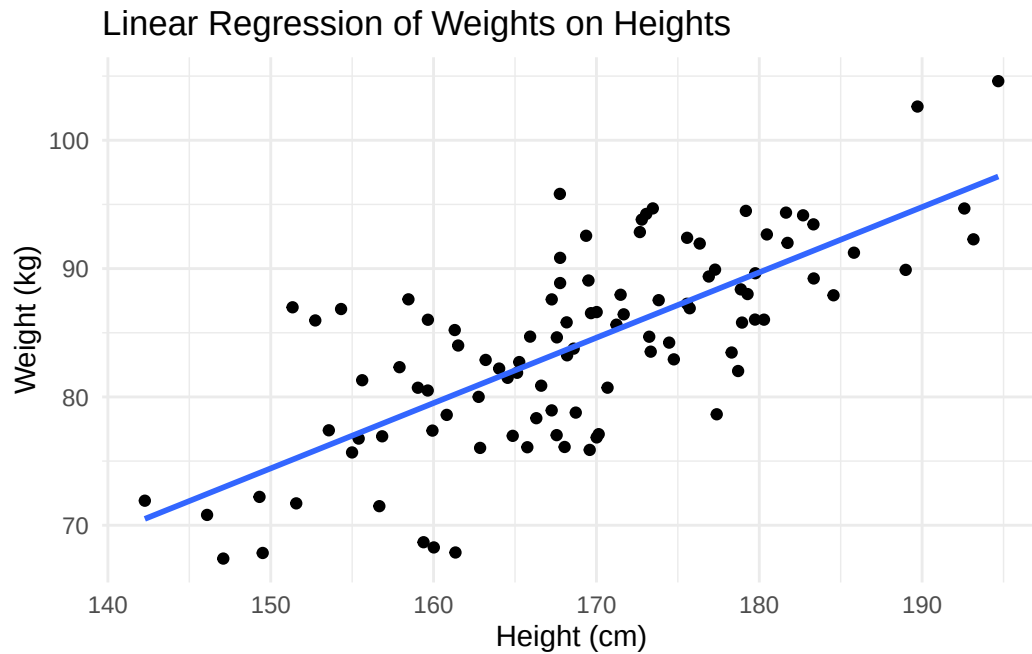
Multiple R-squared: 0.5285, Adjusted R-squared: 0.5237

F-statistic: 109.9 on 1 and 98 DF, p-value: < 2.2e-16

```
# Plot the data and the fitted line
ggplot(df, aes(x = heights, y = weights)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Linear Regression of Weights on Heights",
```

```
x = "Height (cm)",  
y = "Weight (kg)"
```

```
`geom_smooth()` using formula = 'y ~ x'
```



In this example, we generated a data set of heights and weights, fitted a linear regression model to the data, and visualized the relationship between height and weight. This is just a simple example, but it illustrates the types of analyses we will be doing in this course. We will be covering much more complex models and data sets as we progress through the semester.

Throughout the course, we will be using **R** programming software.

1 Matrix Representation and Notation

In this first week, we introduce the mathematical language and statistical framework that will be used throughout the course. Our focus is on the notation of vectors and matrices, vectors of random variables, expectation and covariance operators, and the basic form of the linear regression model.

Learning Objectives

By the end of this week, students should be able to:

- use standard matrix notation for linear statistical models;
- distinguish between scalars, vectors, matrices, random variables, and random vectors;
- compute expectations and covariance matrices for random vectors;
- interpret the linear regression model in matrix form;
- understand why projection and least squares will play a central role in this course.

Reading

Recommended reading for this week:

- Seber and Lee (2003), Chapter 1:
 - 1.1 Notation
 - 1.2 Statistical Models
 - 1.3 Linear Regression Models
 - 1.4 Expectation and Covariance Operators
- Optional preview:
 - Chapter 2: Multivariate Normal Distribution

1.1 Why Linear Statistical Analysis?

Linear statistical analysis is one of the central foundations of graduate statistics. Many methods that at first look different are built on the same underlying structure:

- regression
- analysis of variance
- analysis of covariance
- prediction
- model comparison
- and parts of generalized linear modelling

A major goal of this course is to see these topics under a unified framework.

1.2 A unifying point of view

A large part of the course can be summarized by the model

$$Y = X\beta + \varepsilon,$$

where

- Y is a response vector,
- X is a design matrix,
- β is an unknown parameter vector,
- ε is a random error vector.

This compact expression contains a great deal of statistical structure. Over the semester, we will study how to estimate β , quantify uncertainty, test hypotheses, diagnose model failures, and make predictions.

1.3 Basic Notation

Scalars, vectors, and matrices

We use the following conventions throughout the course:

- scalars are written in lowercase italic letters, such as a , b , n ;
- vectors are written in bold lowercase letters, such as $\mathbf{x}$, $\mathbf{y}$;
- matrices are written in bold uppercase letters, such as $\mathbf{X}$, $\mathbf{A}$;
- random variables are often written in uppercase letters, such as Y ;
- realizations of random variables are written in lowercase letters, such as y .

Univariate case

A random variable is **univariate** if it takes values in one dimension. We write

$$X \in \mathbb{R}.$$

A realization, or observed value, of X is denoted by x .

Some common quantities associated with a univariate random variable X include:

- **Mean**

$$\mu = \mathbb{E}(X).$$

- **Variance**

$$\sigma^2 = \text{Var}(X) = \mathbb{E}[(X - \mu)^2].$$

- **k th moment**

$$\mathbb{E}(X^k).$$

- **Cumulative distribution function (CDF)**

$$F(x) = P(X \leq x).$$

- **Probability density function (PDF)**, for a continuous random variable,

$$f(x).$$

- **Probability mass function (PMF)**, for a discrete random variable,

$$p(x) = P(X = x).$$

- **Quantile**

$$q_\alpha = F^{-1}(\alpha),$$

where $0 < \alpha < 1$.

For a random sample of size n , we write

$$X_1, X_2, \dots, X_n,$$

with observed values

$$x_1, x_2, \dots, x_n.$$

The sample mean is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i,$$

and the sample variance is

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

Multivariate case

A column vector in $\mathbb{R}^n$ is written as

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = (x_1, x_2, \dots, x_n).$$

An $n \times p$ matrix is written as

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix}.$$

Transpose, inverse, and rank

If $\mathbf{A}$ is a matrix, then:

- $\mathbf{A}^\top$ denotes its transpose;
- $\mathbf{A}^{-1}$ denotes its inverse, when it exists;
- $\text{rank}(\mathbf{A})$ denotes its rank;
- $\mathbf{I}_n$ denotes the $n \times n$ identity matrix.

1.3.1 Inner product and norm

For vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the inner product is

$$\mathbf{x}^\top \mathbf{y} = \sum_{i=1}^n x_i y_i.$$

For vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the inner product is

$$\|\mathbf{x}\| = \sqrt{\mathbf{x}^\top \mathbf{x}}.$$

These ideas are fundamental because least squares estimation is based on minimizing squared Euclidean distance.

A random vector is a vector whose entries are random variables. For example,

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}$$

is an n -dimensional random vector.

In linear models, the response is naturally treated as a random vector.

The mean vector of $\mathbf{Y}$ is

$$\mathbb{E}[\mathbf{Y}] = \begin{pmatrix} \mathbb{E}[Y_1] \\ \mathbb{E}[Y_2] \\ \vdots \\ \mathbb{E}[Y_n] \end{pmatrix}.$$

We often write

$$\mathbb{E}[\mathbf{Y}] = \boldsymbol{\mu}.$$

The covariance matrix of $\mathbf{Y}$ is

$$\text{Var}(\mathbf{Y}) = \mathbb{E}[(\mathbf{Y} - \boldsymbol{\mu})(\mathbf{Y} - \boldsymbol{\mu})^\top].$$

Equivalently,

$$\text{Var}(\mathbf{Y}) = \mathbb{E}[\mathbf{Y}\mathbf{Y}^\top] - \mu\mu^\top.$$

If

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \end{pmatrix},$$

then

$$\text{Var}(\mathbf{Y}) = \begin{pmatrix} \text{Var}(Y_1) & \text{Cov}(Y_1, Y_2) & \text{Cov}(Y_1, Y_3) \\ \text{Cov}(Y_2, Y_1) & \text{Var}(Y_2) & \text{Cov}(Y_2, Y_3) \\ \text{Cov}(Y_3, Y_1) & \text{Cov}(Y_3, Y_2) & \text{Var}(Y_3) \end{pmatrix}.$$

1.3.2 Expectation and Covariance Operators

If $\mathbf{A}$ is a constant matrix and $\mathbf{b}$ is a constant vector, then

$$\mathbb{E}[\mathbf{A}\mathbf{Y} + \mathbf{b}] = \mathbf{A}\mathbb{E}[\mathbf{Y}] + \mathbf{b}.$$

This is one of the most useful identities in the course.

If $\mathbf{A}$ is a constant matrix, then

$$\text{Var}(\mathbf{A}\mathbf{Y}) = \mathbf{A} \text{Var}(\mathbf{Y}) \mathbf{A}^\top.$$

More generally, if $\mathbf{Y}$ and $\mathbf{Z}$ are random vectors, then

$$\text{Cov}(\mathbf{A}\mathbf{Y}, \mathbf{B}\mathbf{Z}) = \mathbf{A} \text{Cov}(\mathbf{Y}, \mathbf{Z}) \mathbf{B}^\top.$$

These formulas are essential for deriving the variance of estimators later.

If $Y_1, \dots, Y_n$ are independent and each has variance σ^2 , then

$$\text{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n.$$

This is the most common starting assumption in classical linear regression.

1.4 Statistical Models

A statistical model is a set of probability distributions that may plausibly describe the data-generating mechanism.

1.4.1 General idea

Suppose we observe data y from a random quantity Y . A model introduces assumptions about the distribution of Y , often indexed by an unknown parameter θ .

For example:

$$Y \sim N(\mu, \sigma^2)$$

with unknown parameters μ and σ^2 .

In regression, the model is not only about the marginal distribution of the response but also about how the mean changes with explanatory variables.

1.4.2 Deterministic part and random part

A useful way to think about a statistical model is:

$$\text{data} = \text{systematic part} + \text{random part}.$$

For linear regression, this becomes

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

Here,

- $\beta_0 + \beta_1 x_i$ is the systematic part;
- ε_i is the random part.

1.5 The Linear Regression Model

1.5.1 Simple linear regression

The simplest regression model is

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n.$$

Typical assumptions are

$$\mathbb{E}[\varepsilon_i] = 0, \quad \text{Var}(\varepsilon_i) = \sigma^2, \quad \text{Cov}(\varepsilon_i, \varepsilon_j) = 0 \text{ for } i \neq j.$$

1.5.2 Matrix form

We can write the model compactly as

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon,$$

where

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}, \quad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}, \quad \varepsilon = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

This notation will allow us to treat simple regression, multiple regression, ANOVA, and ANCOVA in one common language.

1.5.3 Mean and variance under the model

If

$$\mathbb{E}[\varepsilon] = \mathbf{0} \quad \text{and} \quad \text{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n,$$

then

$$\mathbb{E}[\mathbf{Y}] = \mathbf{X}\beta$$

and

$$\text{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n.$$

These are immediate consequences of the expectation and covariance rules above.

1.5.4 Geometry Preview

A central idea in linear regression is projection.

The fitted values $\hat{\mathbf{Y}}$ will later be obtained by projecting $\mathbf{Y}$ onto the column space of $\mathbf{X}$.

1.5.5 Column space

The column space of $\mathbf{X}$ is

$$\mathcal{C}(\mathbf{X}) = \{\mathbf{X}\beta : \beta \in \mathbb{R}^p\}.$$

This is the set of all mean vectors that the model can represent.

1.6 Why projection matters

The least squares estimator chooses $\hat{\beta}$ so that

$$\|\mathbf{Y} - \mathbf{X}\beta\|^2$$

is minimized.

So regression is not only algebra. It is also geometry.

Suppose we observe the following data:

x_i	0	1	2	3
y_i	1	3	3	5

Then

$$\mathbf{Y} = \begin{pmatrix} 1 \\ 3 \\ 3 \\ 5 \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}.$$

We will learn next week that the least squares estimator is

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

For now, the main point is to understand how the model is written and how the data are represented in matrix form.

R code

```
x <- c(0, 1, 2, 3)
y <- c(1, 3, 3, 5)

mod <- lm(y ~ x)
summary(mod)
```

Call:

```
lm(formula = y ~ x)
```

Residuals:

```
    1    2    3    4
-0.2  0.6 -0.6  0.2
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.2000	0.5292	2.268	0.1515
x	1.2000	0.2828	4.243	0.0513 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6325 on 2 degrees of freedom

Multiple R-squared: 0.9, Adjusted R-squared: 0.85

F-statistic: 18 on 1 and 2 DF, p-value: 0.05132

```
# Inspecting the design matrix
model.matrix(mod)
```

```

      (Intercept) x
1             1 0
2             1 1
3             1 2
4             1 3
attr(,"assign")
[1] 0 1

```

```

# Fitted values and residuals
fitted(mod)

```

```

      1      2      3      4
1.2 2.4 3.6 4.8

```

```

residuals(mod)

```

```

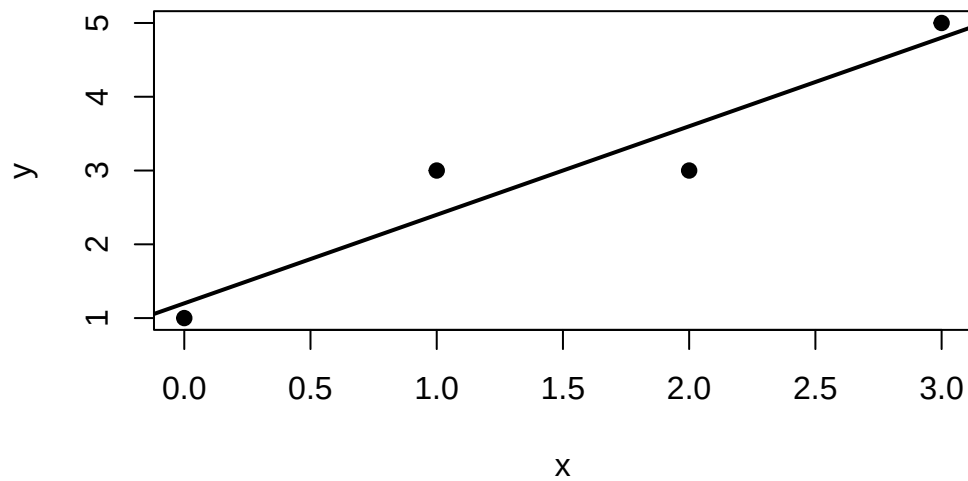
      1      2      3      4
-0.2  0.6 -0.6  0.2

```

```

# quick plot
plot(x, y, pch = 19, xlab = "x", ylab = "y")
abline(mod, lwd = 2)

```



1.7 Discussion

1. Why is it helpful to write regression models in matrix form rather than only scalar notation?
2. What does the covariance matrix tell us that separate variances do not?
3. What is the interpretation of the column space of $\mathbf{X}$?
4. In what sense is regression a projection problem?

1.8 Practice Problems

Conceptual

1. Explain the difference between a random variable and a random vector.
2. Explain why $\text{Var}(\mathbf{Y})$ must be a symmetric matrix.
3. Give an example of a statistical model outside regression.

Computational

Let

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix}$$

with

$$\mathbb{E}[\mathbf{Y}] = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \text{Var}(\mathbf{Y}) = \begin{pmatrix} 4 & 1 & 1 & 9 \end{pmatrix}.$$

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}.$$

Compute:

1. $\mathbb{E}[\mathbf{A}\mathbf{Y} + \mathbf{b}]$
2. $\text{Var}(\mathbf{A}\mathbf{Y})$

Regression setup

For the model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,$$

write out the matrices $\mathbf{Y}$, $\mathbf{X}$, β , and ε for $n = 5$ observations.

1.9 Take home Practice

Complete the following:

- review matrix multiplication and transpose rules;
- derive $\mathbb{E}[\mathbf{A}\mathbf{Y} + \mathbf{b}]$ from first principles;
- derive $\text{Var}(\mathbf{A}\mathbf{Y})$ using the definition of covariance;
- write the simple linear regression model in matrix form for a dataset of your choice;
- fit a simple regression in R and report:
 - the estimated coefficients,
 - fitted values,
 - residuals,
 - and a scatterplot with the fitted line.

1.10 Summary

This week introduced the notation and basic probabilistic tools needed for the rest of the course. We defined random vectors, mean vectors, covariance matrices, and the matrix form of the linear regression model. These ideas will support everything that follows.

Next week, we will study least squares estimation and the geometry of projection in more detail.

For some optional review of the matrix algebra, see [Matrix Algebra Review](#).

References

Montgomery, Douglas C. 2017. *Design and Analysis of Experiments*. John Wiley & Sons.

Montgomery, Douglas C, Elizabeth A Peck, and G Geoffrey Vining. 2021. *Introduction to Linear Regression Analysis*. John Wiley & Sons.

Petersen, Kaare Brandt, and Michael Syskind Pedersen. 2008. *The Matrix Cookbook*. Technical University of Denmark.

Seber, George AF, and Alan J Lee. 2003. *Linear Regression Analysis*. John Wiley & Sons.

Part I

Appendix

Matrix Algebra Review

Matrix dimensions

If $\mathbf{A}$ is $m \times n$ and $\mathbf{B}$ is $n \times p$, then $\mathbf{AB}$ is defined and is $m \times p$.

Transpose rules

For conformable matrices,

$$(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top.$$

Inverse rule

If $\mathbf{A}$ and $\mathbf{B}$ are invertible, then

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1} \mathbf{A}^{-1}.$$

Symmetric matrices

A matrix $\mathbf{A}$ is symmetric if

$$\mathbf{A}^\top = \mathbf{A}.$$

Covariance matrices are always symmetric.

A good reference of the linear algebra can be found from Petersen and Pedersen (2008).

- Petersen, Kaare Brandt, and Michael Syskind Pedersen. (2008). *The Matrix Cookbook*. Technical University of Denmark.